MIL-C-63168(AR)
25 June 1982
SUPERSEDING
(See 6.5)

MILITARY SPECIFICATION

CUSHION, AFT
M753 8 INCH PROJECTILE

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for aft cushions used in M753, 8 Inch Projectiles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

ZZ-R-765	Rubber, Silicone

MILITARY

MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

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STANDARDS

MILITARY

MIL-STD-286	Propellant, Solid Sampling, Examinations and Tests
MIL-STD-1169	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.6)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

11829196	Cushion, aft

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 792	Specific Gravity and Density of Plastics by Displacement
ASTM D 2240	Rubber Property, Durometer Hardness Test for

(Applications for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Assembly. The Aft Cushion shall comply with all requirements specified on drawing (dwg.) 11829196, with all requirements specified herein and with the requirements of all applicable specifications to the extent specified on the dwg. and herein.

3.2 Material. Silicone rubber shall comply with all requirements of FED-SPEC ZZ-R-765 (Class IIb), with all requirements specified herein and with the requirements of all applicable specifications to the extent specified on the dwg. and herein. The contractor shall submit test samples of each lot of material to the procuring agency for analysis and approval in accordance with the requirements of this specification prior to use in fabrication of aft cushions (see 6.3).

3.3 Material properties.

3.3.1 Hardness. The Shore "A" Durometer hardness of the cured silicone rubber shall be 50 ± 5. Testing shall be performed in accordance with 4.5.1.

3.3.2 Density. The density of the cured silicone rubber at 73 ± 3°F (23 ± 3°C) shall be 1.335 ± .013. Testing shall be performed in accordance with 4.5.2.

3.3.3 Composition. The cured silicone rubber shall contain at least 56 percent polymer by weight. Testing shall be performed in accordance with 4.5.3.

3.3.4 Infrared spectrum. The infrared spectrum of the cured silicone rubber shall qualitatively match the wavelength characteristics of Figure 2. Testing shall be performed in accordance with 4.5.4.

3.3.5 Reactivity. The reactivity between the cured silicone rubber and AHH propellant shall not exceed 1.0 milliliter (ml) when tested in accordance with 4.5.5.

3.4 First article. This specification contains technical provisions for first article inspection (see 4.3). Requirements for submission of first article samples shall be as specified in the contract.

3.5 Workmanship. Cushions shall be fabricated in a thoroughly workmanlike manner and shall be free of large voids or any other obvious material defects. All finished parts shall be clean and free of chips, dirt, grease, contamination or any other foreign matter. Cleaning agents

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shall not be injurious to the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract or purchase order, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. When the contract requires first article inspection (see 3.4), the contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with 4.3.2. The sample shall consist of the following items in the sample quantities as indicated.

Item	Drawing No.	Quantity
Aft Cushions	11829196	6
Specimen of cured silicon rubber for material analysis (see 3.2)	11829196	1 square foot

4.3.2 Inspections to be performed. The first article samples submitted in accordance with 4.3.1 shall be subjected to any or all of the requirements of the applicable dwg. and to the inspections and tests specified in the first article inspection table herein. All parts shall have been manufactured using the same production processes as will be used

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in fulfilling the contract and all materials shall be obtained from the same source as will be used in regular production and approved in accordance with 3.2. If the first article sample fails to meet any of the applicable requirements, the first article sample shall be rejected.

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FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63168(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.1	Cushion, Aft				11829196
					NEXT HIGHER ASSEMBLY
					M753 Proj.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Material Properties	(a)			4.4.2.1
	Classification of Defects and Tests	6			4.4.2.2

NOTES: (a) One square foot of cured silicone rubber (and a certificate of conformance for material per FED-SPEC ZZ-R-765 (Class IIb)) shall be submitted to the Government (see 6.3) for material property test approval prior to use in fabrication of cushions.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots of finished cushions shall comply with the lot formation provisions of MIL-A-48078. In addition, an inspection lot shall consist of items fabricated from a single inspection lot of material produced by a single manufacturer at one location in one continuous process employing not more than one lot of base material and curing agent without change in formulation or process and approved in accordance with 3.2.

4.4.2 Examinations and tests. See MIL-A-48078 and the Quality Conformance Inspection tables herein. If any sample or specimen fails to meet any of the applicable requirements, the inspection lot which it represents shall be rejected.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH 4.4.2.1	TITLE Material Properties of Silicon Rubber		SHEET 1 OF 1		DRAWING NUMBER 11829196
					NEXT HIGHER ASSEMBLY M753 Proj.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Shore "A" Durometer hardness	(a)		3.3.1	4.5.1
102.	Density	(a)		3.3.2	4.5.2
103.	Composition	(a)		3.3.3	4.5.3
104.	Infrared spectrum	(a)		3.3.4	4.5.4
105.	Reactivity	(a)		3.3.5	4.5.5

NOTES: (a) One square foot of cured silicon rubber (and a certificate of conformance for material per FED-SPEC ZZ-R-765 (Class IIb)) shall be submitted to the Gov't (see 6.3) for material property test approval prior to use in fabrication of cushions.

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QUALITY CONFORMANCE INSPECTION

MIL-C-63168(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Cushion, Aft	1 OF 1	11829196
			NEXT HIGHER ASSEMBLY: M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Outside diameter	4.4. 3.1		3.1	Gage/TMC
102.	Inside diameter	4.4. 3.1		3.1	Gage/TMC
103.	Location inside diameter	4.4 3.1		3.1	Gage/TMC
104.	Thickness	(a)		3.1	Gage
105.	Part identification		100%	3.1	Visual
Minor:					
201.	Poor workmanship		100%	3.5	Visual

NOTES: (a) Gage each sheet of material when sheet material is used or one of every ten cushions if other fabrication methods are used.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63168(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Packaging	1 OF 1	See 5.0
			NEXT HIGHER ASSEMBLY: N.A.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Packaging and packing not in accordance with applicable requirements		-	5.0	Visual
102.	Items not clean and free of contamination and foreign material prior to packaging		100%	5.1.1	Visual
103.	Fiberboard separators missing		100%	5.1.1	Visual
104.	Spiral container improper or improperly sealed		100%	5.1.1	Visual
105.	Filler and cushioning does not provide firm pack		100%	5.1.1	Visual/Manual
106.	Exterior pack improper or not properly sealed		100%	5.1.2	Visual
107.	Marking on spiral wound or exterior containers missing, improper or illegible		100%	5.1.3	Visual
108.	Evidence of poor workmanship		100%	-----	Visual

NOTES:

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4.4.3 Testing. Testing is described in the first article and quality conformance inspection tables and herein.

4.4.3.1 Tool manufacturing control. When tool manufacturing control (TMC) is specified, the first five pieces initially produced from each mold or die inspected for the applicable classification of defects herein shall constitute initial tool acceptance. Subsequently, the first and last five pieces from each days production inspected for the applicable classification of defects herein shall constitute inspection lot acceptance. In addition, any time the tool requires repair an additional five pieces from each tool shall be inspected for the applicable classification of defects herein before resumption of production. If any sample fails to meet any of the applicable inspection requirements, the lot of production under consideration shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" columns in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see section 6 of MIL-A-48078 and 6.4 herein).

4.5 Test methods and procedures.

4.5.1 Hardness (see 3.3.1). The Shore "A" Durometer hardness of the material shall be measured in accordance with ASTM D2240 or an approved equivalent.

4.5.2 Density (see 3.3.2). The density of the silicone rubber shall be determined in accordance with ASTM D792, Method 1.

4.5.3 Composition (see 3.3.3). The polymer content of the silicone rubber shall be determined using a DuPont 950 Thermogravimetric Analyzer or an approved equivalent to measure and record the sample weight loss as a function of temperature as shown in Figure 1. The rate of temperature change shall be $5^{o}C$ per minute and the medium shall be nitrogen.

4.5.4 Infrared spectrum (see 3.3.4). The infrared spectrum of the silicone rubber shall be determined by pyrolyzing a sample in a test tube over a bunsen burner, transferring the resultant liquid condensate into an infrared transmitting window and recording the infrared spectrum with an infrared spectrophotometer.

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4.5.5 Reactivity (see 3.3.5). Reactivity tests shall be performed in accordance with MIL-STD-286, Method 405.1.

5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified in the contract or order, packaging and packing shall be in accordance with MIL-STD-1169 subject to the following details and conditions.

5.1.1 Packaging. Items shall be semi-bulk packaged in moistureproof spirally wound containers in accordance with MIL-STD-1169, Level A. Items shall be separated by fiberboard separators and fillers and cushioning provided as required.

5.1.2 Packing. Spiral wound containers shall be packed in accordance with MIL-STD-1169, Level B.

5.1.3 Marking. Spiral wound containers and exterior containers shall be marked with the following:

a. Item nomenclature
b. Part or dwg. no
c. Used on M753 Projectile
d. Manufacturer
e. Quantity inclosed
f. Lot no.
g. Contract or purchase order no.
h. Date of acceptance

6. NOTES

6.1 Intended use. The aft cushions covered by this specification are intended for use in assembly of rocket motors for M753, 8 Inch Projectiles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. First article inspection, when required (see 3.4).

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirements	Applicable DID no.
4.4.4	Inspection equipment approval	DI-R-1714 (Tailored)

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Test samples. Material samples shall be submitted for approval to Commander U.S. Army Armament Research and Development Command, ATTN: DRDAR-LCA-OP, Dover, New Jersey 07801 or an approved laboratory.

6.4 Equipment design approval. Submit equipment designs, as required, to Commander U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.5 Supersession data. This specification includes the requirements of Purchase Description AR-PD-13 dated 9 April 1979.

6.6 Drawings and specifications. Drawings and specifications listed in Section 2 of this specification under the heading U.S. Army Armament Research Development Command may also include dwgs. and specifications prepared and identified as Frankford Arsenal, Edgewood Arsenal, Rock Island Arsenal or Picatinny Arsenal dwgs. and specifications. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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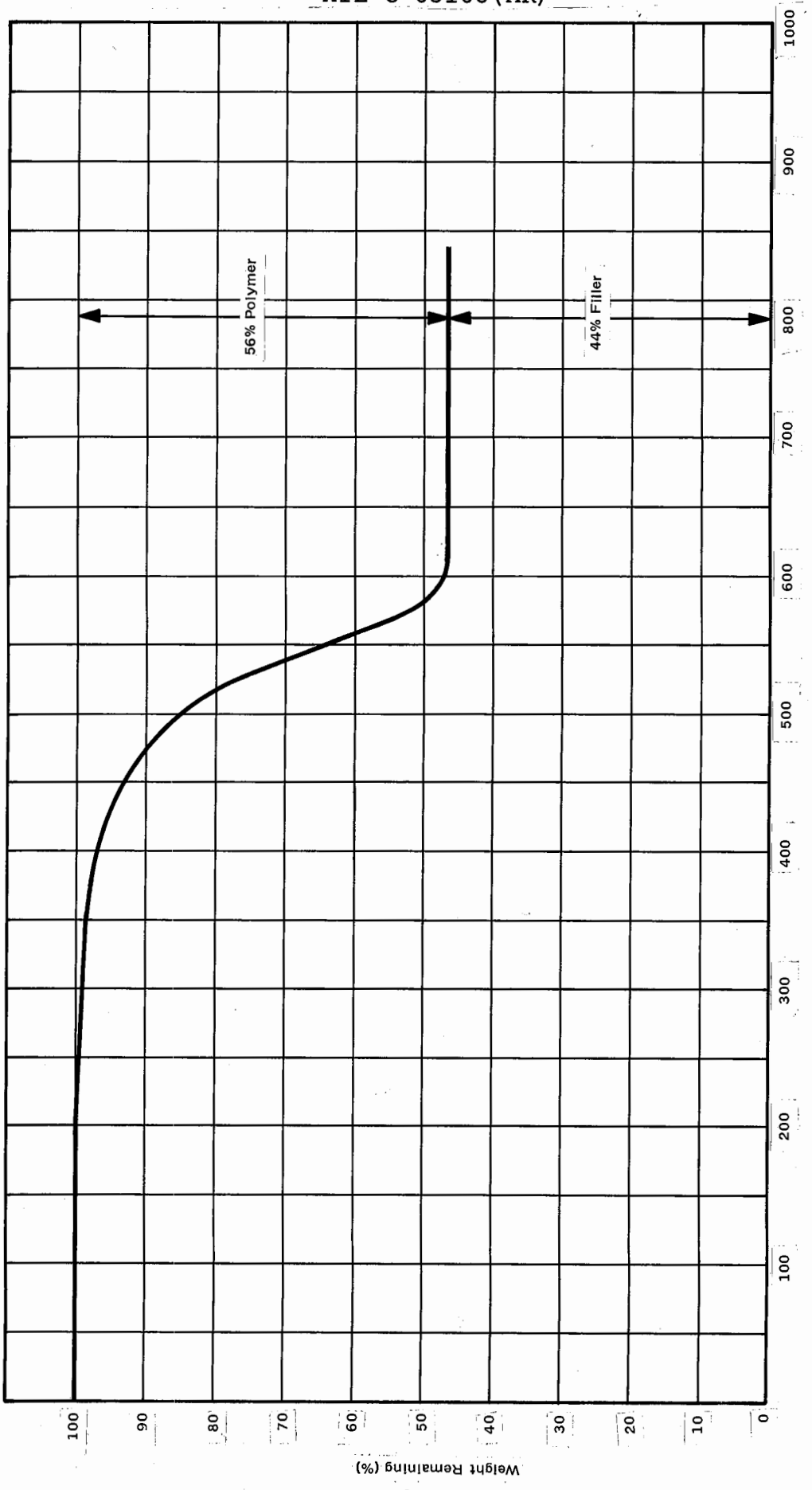


FIGURE 1. Typical thermogravimetric analysis of weight loss.

Figure 1. Typical thermogravimetric analysis of weight loss.

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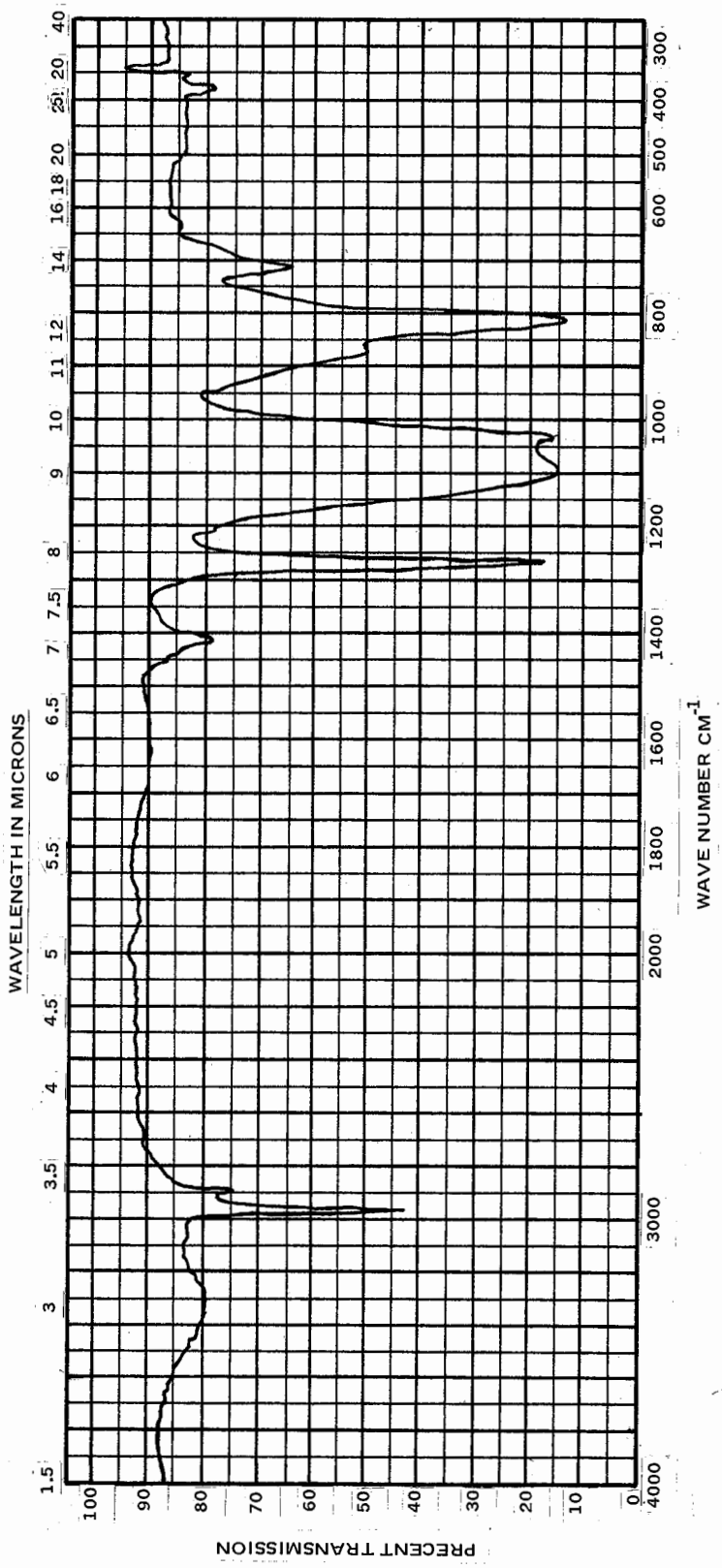


FIGURE 2. Infrared spectrum.

Figure 2. Infrared spectrum.

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